UNIT 5 · 8 hours

State-Space Model Using the Physical Variable Approach (SISO)



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

U5

Learning objectives

- Choose physical variables as state variables
- Derive state-space models of RLC and mass–spring–damper systems
- Write the A, B, C, D matrices
- Check using the transfer function

Syllabus context

Introduction to state variable analysis: modelling of physical systems using the state-space approach, advantages of state space over the transfer function approach, state-space model using the physical variable approach for SISO systems.

U5

Physical variable approach

1

Identify storage

Find independent energy-storing elements (L, C, M, K, J).

2

Choose states

Inductor current, capacitor voltage; mass position and velocity; etc.

3

Write laws

KVL/KCL or Newton's law for each storage element: first-order equations.

4

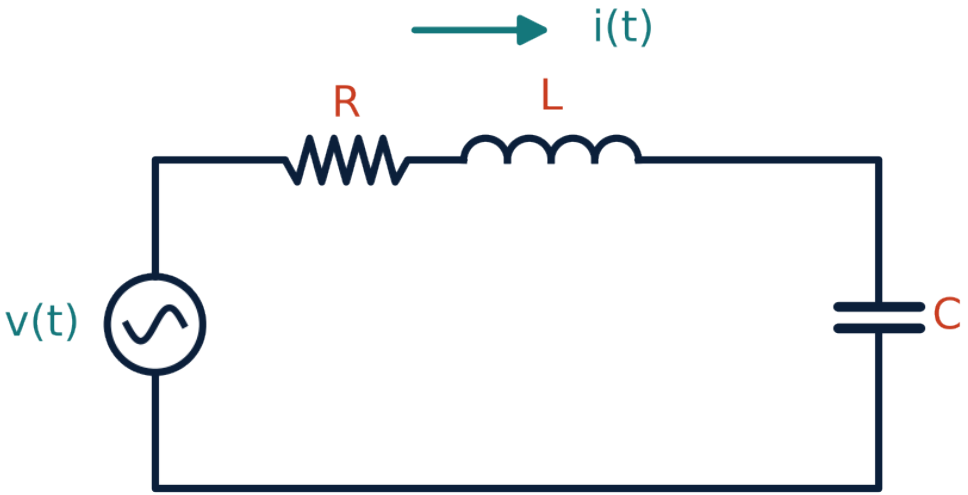
Matrix form

Collect into $\dot{x} = Ax + Bu$, $y = Cx$.

Number of states = number of independent energy storage elements.

U5

Series RLC circuit



- States: x1 = i (inductor current), x2 = vc (capacitor voltage).
- L di/dt = u − R i − vc.
- C dvc/dt = i.
- Output y = vc.

U5

RLC state model

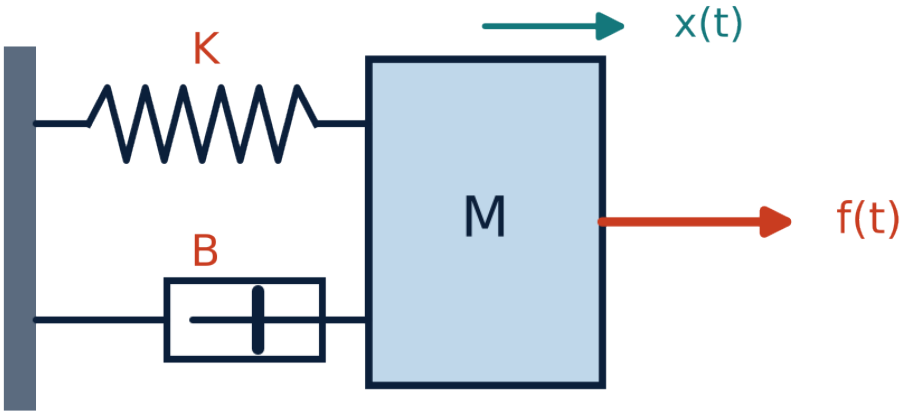
Problem Write the state model for R = 3 Ω, L = 1 H, C = 0.5 F (input u = source voltage, output vc). Verify with the transfer function.

- $\dot{x}1 = -(R/L)x1 - (1/L)x2 + (1/L)u = -3x1 - x2 + u$
- $\dot{x}2 = (1/C)x1 = 2x1$
- $A = [-3\ -1;\ 2\ 0],\ B = [1;\ 0],\ C = [0\ 1],\ D = 0$
- $\det(sI - A) = s^2 + 3s + 2;\ \ G = C(sI - A)^{-1}B = 2/(s^2 + 3s + 2)$

Answer: Eigenvalues −1, −2; $G(s) = 2/(s^2 + 3s + 2)$

U5

Mass–spring–damper



- States: x1 = x (position), x2 = ẋ (velocity).
- ẋ1 = x2
- ẋ2 = −(K/M)x1 − (B/M)x2 + (1/M)f
- Output y = x1.

U5

Mass–spring–damper state model

Problem M = 1 kg, B = 3 N·s/m, K = 2 N/m. Write A, B, C and check the poles.

- A = [0 1; −K/M −B/M] = [0 1; −2 −3]
- B = [0; 1/M] = [0; 1], C = [1 0], D = 0
- $\det(sI - A) = s^2 + 3s + 2 \Rightarrow$ eigenvalues −1, −2
- $G(s) = 1/(s^2 + 3s + 2)$ — same as Unit 1

Answer: A = [0 1; −2 −3], B = [0; 1], C = [1 0]

U5

Remarks

- State-space model is not unique: a different state choice gives a different (similar) A.
- Phase variable form can be derived directly from the transfer function.
- Physical variables preferred when the internal quantities are of interest or must be measured.
- MATLAB: sys = ss(A, B, C, D); step(sys), eig(A), tf(sys).

U5

Key takeaways

- Choose physical energy-storage variables as states
- RLC: x = [i, vc]; mass-spring: x = [position, velocity]
- $\dot{x} = Ax + Bu$, $y = Cx$; poles = eig(A)
- Verify with $G(s) = C(sI - A)^{-1}B$

U5

Practice questions

Q1. Derive the state model of a series RLC circuit.

Q2. Write A, B, C for M = 2, B = 4, K = 8.

Q3. Find the eigenvalues of A = [0 1; −6 −5].

Q4. Convert the state model to a transfer function.